

Meeting 02 - Outline

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From Food Belt to Migration Belt:

Assessing the Socio-Spatial Impacts of the 2010 Drought on Agricultural Land Use and Rural Migration in Guizhou, China

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Zheng et al. (2024)

Wang et al. (2024)

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Yun et al. (2012)

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3. Spatial Heterogeneity Assessment:

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Introduction

Key words: Extreme climate shock | Agricultural fragility | Rural out-migration | Socio-spatial mechanisms

Background

In 2010, one of the worst droughts in a century hit south-west China, hitting the fragile karst agricultural province of Guizhou. The drought devastated the livelihoods of local populations, ecosystems and rural production systems.

Historically part of China's 'food security belt', Guizhou Province now faces increasing ecological pressures, making it an important area for studying climate-agriculture-migration linkages.

Research Problem

Extreme weather events are expected to exacerbate agricultural stress and rural exodus. However, empirical socio-spatial mechanisms in less developed mountain regions remain under-explored.

Research Aim & Questions:

This study aims to investigate whether and how the 2010 drought in Guizhou affected agricultural viability and prompted rural labour migration.

Key questions include:

Did the drought significantly reduce agricultural productivity (measured via NDVI)?

Was there a corresponding increase in rural out-migration?

Were these effects spatially heterogeneous across counties?

Motivation

This study bridges the gap between environmental shocks, agricultural land use change and labour mobility in understudied areas of inland China, providing evidence for adaptation strategies.

Literature Review

Key words: Income–migration mechanism | Spatial heterogeneity | Remote sensing | Drought-specific analysis

Zheng et al. (2024)

This study uses panel data from over 2,900 rural households across China to examine how climate variability —especially changes in temperature and precipitation —affects agricultural labor allocation. The authors find that reductions in net agricultural income, driven by climate anomalies, significantly push farmers toward non—agricultural employment, particularly those with lower risk tolerance.

Gap: While the paper provides rigorous micro-level insights into income-induced migration, it lacks a spatially explicit or event-based framework. The findings are not contextualised around specific climate shocks such as droughts, nor do they address spatial heterogeneity across regions like mountainous Guizhou.

Wang et al. (2024)

Using longitudinal survey and precipitation index (SPI) data, this article investigates how extreme rainfall events shape household agricultural yields, income levels, and labour mobility in rural China. It highlights that households in less affluent areas are more vulnerable, with a higher likelihood of temporary migration following shocks.

Gap: The study primarily focuses on rainfall (not drought) and samples heavily from more economically developed eastern provinces. There is limited analysis of ecologically fragile, drought-prone, or mountainous regions such as southwest China, where migration patterns and environmental responses may differ.

Liang et al. (2021)

This research leverages Landsat satellite imagery to map the northward shift of the northern limit of paddy rice cultivation between 1984 and 2013. By integrating land use data with temperature trends, the authors establish a clear link between climate change and shifts in cultivation patterns, mediated by elevation and regional policy factors.

Gap: While the study presents a robust approach to analyzing long-term vegetation-climate interactions using remote sensing, it focuses on gradual climate trends (e.g., warming), rather than discrete shocks like droughts. It also does not explore the socio-demographic consequences of land productivity change.

Yun et al. (2012)

This case study examines social perceptions and responses to the 2009–2010 drought in Qianxinan Prefecture, Guizhou. Drawing on field surveys, meteorological indices, and media reports, it reconstructs the evolution of drought cognition and policy response. The study reveals that households detect drought impacts earlier than institutions, and that social responses evolve from individual to institutional levels.

Gap: Although contextually rich and focused on the precise study region, the paper lacks a quantitative evaluation of land or population change. It also does not incorporate geospatial or remote sensing data to validate environmental impact.

Study's Contribution:

By combining county-level remote sensing (NDVI), climate indices (SPEI), and demographic data, this study employs a quasi-experimental design to assess the spatially differentiated impacts of drought on land and population, addressing a regional empirical gap in the climate-induced migration literature.

Materials and Methods

Study Area:

Focus: Guizhou Province, with particular attention to Qianxinan, Bijie, and Liupanshui—areas heavily impacted by the 2010 drought.

Reason: Karst terrain, agricultural dependence, and high poverty rates make this an ideal site to observe climate vulnerability.

Data Sources:

- **Climate:**
 - CHIRPS rainfall and ERA5-Land temperature data
 - SPEI index for drought quantification (2008–2012)
- **Agriculture & Land:**
 - MODIS NDVI (MOD13Q1) to track vegetation dynamics
 - MODIS Land Cover (MCD12Q1) or ESA WorldCover for rural land classification
- **Population:**
 - Guizhou statistical yearbooks and population census (2000, 2010)
 - Migration or labour force data by county, where available

Methodology Framework:

1. Event Study + DiD Design:

Use 2010 as a treatment year

Compare pre- and post-NDVI and population change in severely vs. mildly affected counties

Control for fixed effects and time trends

2. Temporal NDVI Analysis:

Extract NDVI time series (2008–2012) at county or 250m grid level

Measure drops in vegetation vitality as proxy for agricultural stress

3. Spatial Heterogeneity Assessment:

Map and cluster counties with most significant NDVI and demographic shifts

Supplement with policy reports or qualitative sources for triangulation

Ethics Consideration:

All data are anonymized and public; no human subjects involved. A Statement of Ethics will be provided.

Results / Discussion / Conclusion

Key words: Agricultural stress | Migration trigger | Spatial disparity | Policy relevance

Anticipated finding:

Severe drought counties show a drop in NDVI and population retention, suggesting a causal link between climatic shock and rural out-migration.

Discussion will relate to:

Spatial inequality in drought impact

Climate adaptation in ecologically fragile zones

Implications for rural revitalisation policy and land-use planning in western China